

Finding the mean of grouped data

Grouped data is data that has been collected into groups of values, rather than appearing as specific or individual values. If a frequency table shows grouped data, there is not enough information to calculate the sum of values, so only an estimated value for the mean can be found.

$$\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$$

estimated average size of data (pointing to Mean)
total frequency $\times$ midpoint value (pointing to Sum of values)
total frequency (pointing to Number of values)

In grouped data the sum of the values can be calculated by finding the midpoint of each group and multiplying it by the frequency. Then add the results for each group together to find the total frequency $\times$ midpoint value. This is divided by the total number of values to find the mean. The example below shows a group of marks scored in a test.

Mark	Frequency
under 50	2
50–59	1
60–69	8
70–79	5
80–89	3
90–99	1

Mark	Frequency	Midpoint	Frequency $\times$ midpoint
under 50	2	25	$2 \times 25 = 50$
50–59	1	54.5	$1 \times 54.5 = 54.5$
60–69	8	64.5	$8 \times 64.5 = 516$
70–79	5	74.5	$5 \times 74.5 = 372.5$
80–89	3	84.5	$3 \times 84.5 = 253.5$
90–99	1	94.5	$1 \times 94.5 = 94.5$

total frequency

total frequency $\times$ midpoint

To find the midpoint of a set of data, add the upper and lower values and divide the answer by 2. For example, the midpoint in the 90–99 mark group is 94.5.

Multiply the midpoint by the frequency for each group and enter this in a new column. Add the results to find the total frequency multiplied by the midpoint.

Dividing the total frequency $\times$ midpoint by the total frequency gives the estimated mean mark. It is an estimated value as the exact marks scored are not known – only a range has been given in each group.

LOOKING CLOSER

The modal class

In a frequency table with grouped data, it is not possible to find the mode (the value that occurs most often in a group). But it is easy to see the group with the highest frequency in it. This group is known as the modal class.

▷ **More than one modal class**
When the highest frequency in the table is in more than one group, there is more than one modal class.

Mark	0–25	26–50	51–75	76–100
Frequency	2	6	8	8

modal class

LOOKING CLOSER

Weighted mean

If some individual values within grouped data contribute more to the mean than other individual values in the group, a “weighted” mean results.

Students in group	15	20	22
Mean exam mark	18	17	13

$$\frac{(15 \times 18) + (20 \times 17) + (22 \times 13)}{15 + 20 + 22} = 15.72$$

students $\times$ mean (pointing to 15 $\times$ 18)
students $\times$ mean (pointing to 20 $\times$ 17)
students $\times$ mean (pointing to 22 $\times$ 13)
sum of these three values is total number students (pointing to denominator)
weighted mean (pointing to 15.72)

△ Finding the weighted mean

Multiply the number of students in each group by the mean mark and add the results. Divide by the total students to give the weighted mean.